

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Limits

Week 5 Day 1 AM



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

COURSE SURVEY

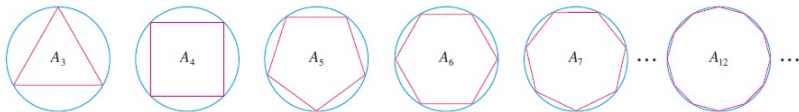
The SSILP Mathematics Survey is now open until 10 Aug. 2026 (11.59pm), you should have received an email on this. Please follow the link provided in the email to complete the survey. Your feedback is very important for us to improve the course.



We have to make this intuition rigorous, which can be done via the idea of limits. To understand limits, it's easier to start with the simplest case, the limits of sequences.

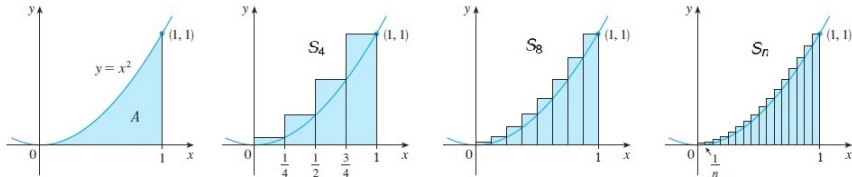
MOTIVATION

The idea of limits of sequences was used by the ancient Greeks to find the area of a circle.



$$\text{Area of a circle} = \lim_{n \rightarrow \infty} \text{Area of inscribed regular } n\text{-gon} = \lim_{n \rightarrow \infty} A_n.$$

This idea is also used to find the area under curves, which you'll see in integral calculus.



$$A = \lim_{n \rightarrow \infty} S_n.$$

CONVERGENCE AND DIVERGENCE

However, there are sequences that we can't tell intuitively whether they converge or diverge, or what value do they converge to.

E.g.: The sequence of the side length of an n -dimensional 'cube' with n -dimensional volume $= n$:

$$\left(\sqrt[n]{n}\right)_{n=1}^{\infty} = \left(1, \sqrt{2}, \sqrt[3]{3}, \sqrt[4]{4}, \dots, \sqrt[n]{n}, \dots\right).$$

E.g.: A sequence related to compound interest that we saw in Week 2:

$$\left(\left(1 + \frac{1}{n}\right)^n\right)_{n=1}^{\infty} = \left(2, \frac{9}{4}, \frac{64}{27}, \dots, \left(1 + \frac{1}{n}\right)^n, \dots\right).$$

Hence, we have a need to define more carefully what it means for a sequence to converge.

DEFINITION OF LIMIT AND CONVERGENT SEQUENCES

DEFINITION[†]:

Intuitively, we say that a sequence $(x_n)_{n=1}^{\infty}$ **converges to** $L \in \mathbb{R}$ and denote this as

$$\lim_{n \rightarrow \infty} x_n = L,$$

if the distance between x_n and L is arbitrarily small whenever n is large enough. In this case, the sequence is called **convergent** and the (unique) number L is called the **limit** of the sequence.

[†] This is not the actual definition. The actual definition is technical and abstract. It's called the ε - N definition of limit.

See this [Geogebra applet](#).

E.g.: $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ and $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1.$

DEFINITION OF DIVERGENT SEQUENCES

DEFINITION:

If a sequence does not converge to **any** $L \in \mathbb{R}$, we say that the sequence **diverges** and it is called **divergent**.

E.g.: The sequence $((-1)^n)_{n=0}^{\infty}$ diverges.

E.g.: The sequence $(n)_{n=1}^{\infty}$ diverges.

DEFINITION:

If the term x_n in a sequence gets arbitrarily large (resp. negative large) as n tends to infinity, we say that the sequence **diverges to ∞ (resp. $-\infty$)** and denote this as

$$\lim_{n \rightarrow \infty} x_n = \infty.$$

$$(\text{resp. } \lim_{n \rightarrow \infty} x_n = -\infty.)$$

E.g.: $\lim_{n \rightarrow \infty} n^2 = \infty$ and $\lim_{n \rightarrow \infty} (1 - 2n) = -\infty.$

ACTIVITY 1

By using this [Geogebra applet](#), find the limits L of the following sequences (if they converge).

(a) $\left(\frac{n^2-3n+7}{2n^2+1}\right)_{n=1}^{\infty}$

(b) $\left(\frac{2^n}{n^{20}}\right)_{n=1}^{\infty}$

(c) $\left(\left(1 + \frac{1}{n}\right)^n\right)_{n=1}^{\infty}$

ACTIVITY 1: SOLUTION

(a)

$$\lim_{n \rightarrow \infty} \frac{n^2 - 3n + 7}{2n^2 + 1} = 0.5.$$

(b)

$$\lim_{n \rightarrow \infty} \frac{2^n}{n^{20}} = \infty.$$

We need to plot the points for $n \geq 140$ to see the sequence diverges to infinity.

(c)

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \approx 2.72.$$

We saw this before in Week 2.

DISADVANTAGES OF THE GRAPHICAL METHOD

What are the disadvantages of the graphical method for finding the limit of a sequence?

- We need to plot the graph.
- The graph can be misleading as it only plots a finite number of points. (e.g. sequence (b))
- Even if the y values of the points clearly tend towards a certain limit, it's hard to find out precisely what that value is. (e.g. sequence (c))

That's why mathematicians have to define the limit of a sequence rigorously and not leave it to our intuition. The ε - N definition of limit is more rigorous but it's quite technical and abstract for non-math major students. Thus, we'll be content with computing limits using rules and theorems.

RULES OF COMPUTING LIMITS

THEOREM

Suppose $\lim_{n \rightarrow \infty} a_n = L$ (L can be $\pm\infty$) and let $c \in \mathbb{R}$ be a constant. Then

- $\lim_{n \rightarrow \infty} c = c,$
- $\lim_{n \rightarrow \infty} ca_n = c \left(\lim_{n \rightarrow \infty} a_n \right) = cL,$

with the conventions ' $\infty \times c = \infty$ ' if $c > 0$, ' $\infty \times c = -\infty$ ' if $c < 0$.

E.g.: $\lim_{n \rightarrow \infty} \pi = \pi.$

E.g.: $\lim_{n \rightarrow \infty} \frac{-3}{n} = -3 \times \lim_{n \rightarrow \infty} \frac{1}{n} = -3 \times 0 = 0.$

E.g.: $\lim_{n \rightarrow \infty} (-2n\pi) = (-2\pi) \lim_{n \rightarrow \infty} n = (-2\pi) \times \infty = -\infty.$

RULES OF COMPUTING LIMITS

THEOREM

Suppose $\lim_{n \rightarrow \infty} a_n = L_1$ and $\lim_{n \rightarrow \infty} b_n = L_2$ (L_1, L_2 can be $\pm\infty$).
Then

$$\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n = L_1 \pm L_2,$$

with the conventions

- ' $\infty \pm c = \infty$ ', ' $-\infty \pm c = -\infty$ ' for any finite $c \in \mathbb{R}$.
- ' $\infty + \infty = \infty$ '
- ' $-\infty - \infty = -\infty$ '

⚠ ' $\infty - \infty$ ' is an **indeterminate form**, i.e. you **can't make a conclusion** and a different approach is needed.

E.g.: $\lim_{n \rightarrow \infty} \left(2 - \frac{1}{n}\right) = \lim_{n \rightarrow \infty} 2 - \lim_{n \rightarrow \infty} \frac{1}{n} = 2 - 0 = 2.$

E.g.: $\lim_{n \rightarrow \infty} (n - n^2) = \lim_{n \rightarrow \infty} n - \lim_{n \rightarrow \infty} n^2 = \infty - \infty$, an indeterminate form. We need a different approach to compute this limit (see next page).

RULES OF COMPUTING LIMITS

THEOREM

Suppose $\lim_{n \rightarrow \infty} a_n = L_1$ and $\lim_{n \rightarrow \infty} b_n = L_2$ (L_1, L_2 can be $\pm\infty$).
Then

- $\lim_{n \rightarrow \infty} (a_n b_n) = \lim_{n \rightarrow \infty} a_n \times \lim_{n \rightarrow \infty} b_n = L_1 L_2,$
- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} = \frac{L_1}{L_2},$

with the conventions ' $\infty \times \infty = \infty$ ', ' $(-\infty) \times (-\infty) = \infty$ ',
' $\infty \times (-\infty) = -\infty$ ', ' $\frac{1}{\pm\infty} = 0$ ', ' $\frac{1}{0^+} = \infty$ ', ' $\frac{1}{0^-} = -\infty$ '.

⚠ ' $\pm\infty \times 0$ ', ' $\frac{0}{0}$ ', ' $\frac{\infty}{\infty}$ ' are **indeterminate forms**, i.e. you **can't make a conclusion** and a different approach is needed.

E.g.: $\lim_{n \rightarrow \infty} (n - n^2) = \lim_{n \rightarrow \infty} n(1 - n) = \lim_{n \rightarrow \infty} n \times \lim_{n \rightarrow \infty} (1 - n) = \infty \times (-\infty) = -\infty.$

E.g.: $\lim_{n \rightarrow \infty} \left(\frac{8}{n - n^2} \right) = \frac{8}{\lim_{n \rightarrow \infty} n - n^2} = 8 \times \frac{1}{-\infty} = 8 \times 0 = 0.$

COMBINING THE RULES - EXAMPLE

E.g.:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} (-n^2 + 100n + 2024) &= \lim_{n \rightarrow \infty} n^2 \left(-1 + \frac{100}{n} + \frac{2024}{n^2} \right) \\
 &= \lim_{n \rightarrow \infty} n^2 \times \lim_{n \rightarrow \infty} \left(-1 + \frac{100}{n} + \frac{2024}{n^2} \right) \\
 &= \infty \times \left(\lim_{n \rightarrow \infty} -1 + \lim_{n \rightarrow \infty} \frac{100}{n} + \lim_{n \rightarrow \infty} \frac{2024}{n^2} \right) \\
 &= \infty \times (-1 + 0 + 0) = -\infty.
 \end{aligned}$$

If we proceed the following way:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} (-n^2 + 100n + 2024) &= \lim_{n \rightarrow \infty} (-n^2) + \lim_{n \rightarrow \infty} 100n + \lim_{n \rightarrow \infty} 2024 \\
 &= -\infty + \infty + 2024,
 \end{aligned}$$

we cannot conclude as $-\infty + \infty$ is an indeterminate form.

COMBINING THE RULES - EXAMPLE

E.g.:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{n^2 - 3n + 7}{2n^2 + 1} &= \lim_{n \rightarrow \infty} \frac{n^2 \left(1 - \frac{3}{n} + \frac{7}{n^2}\right)}{n^2 \left(2 + \frac{1}{n^2}\right)} \\
 &= \frac{\lim_{n \rightarrow \infty} \left(1 - \frac{3}{n} + \frac{7}{n^2}\right)}{\lim_{n \rightarrow \infty} \left(2 + \frac{1}{n^2}\right)} \\
 &= \frac{\lim_{n \rightarrow \infty} 1 - \lim_{n \rightarrow \infty} \frac{3}{n} + \lim_{n \rightarrow \infty} \frac{7}{n^2}}{\lim_{n \rightarrow \infty} 2 + \lim_{n \rightarrow \infty} \frac{1}{n^2}} \\
 &= \frac{1 - 0 + 0}{2 + 0} = \frac{1}{2}.
 \end{aligned}$$

Can you identify the rules used in each of the examples above?

There's no standard recipe for finding limits of sequences, but one trick that you can try is to factor out the dominant term, as shown in the last two examples.

ACTIVITY 2

Find the limits of the sequences, if the limits exist.

(a) $\lim_{n \rightarrow \infty} \frac{n^2}{n + 100}$

(b) $\lim_{n \rightarrow \infty} \left(\frac{1}{n} - \frac{1}{n + 1} \right)$

(c) $\lim_{n \rightarrow \infty} (1 + (-1)^n)$

ACTIVITY 2: SOLUTION

$$(a) \lim_{n \rightarrow \infty} \frac{n^2}{n+100} = \lim_{n \rightarrow \infty} \frac{n}{1 + \frac{100}{n}} = \frac{\infty}{1+0} = \infty.$$

$$(b) \lim_{n \rightarrow \infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} \frac{n+1-n}{n(n+1)} = \lim_{n \rightarrow \infty} \frac{1}{n(n+1)} = \frac{1}{\infty} = 0.$$

Alternatively,

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} - \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 - 0 = 0.$$

(c) Since $(1 + (-1)^n)_{n=1}^{\infty} = (0, 2, 0, 2, \dots)$ is oscillating between 0 and 2, the sequence diverges.

THEOREMS OF COMPUTING LIMITS

THEOREM

Let f be a continuous function (e.g. a rational power of a rational function, an exponential function, a logarithmic function or a trigonometric function). If $\lim_{n \rightarrow \infty} a_n = L$ and L is in the domain of f , then

$$\lim_{n \rightarrow \infty} f(a_n) = f\left(\lim_{n \rightarrow \infty} a_n\right) = f(L).$$

THEOREM

Let r be a constant. Then

- If $r > 1$, $\lim_{n \rightarrow \infty} r^n = \infty$.
- If $r = 1$, $\lim_{n \rightarrow \infty} r^n = 1$.
- If $|r| < 1$, $\lim_{n \rightarrow \infty} r^n = 0$.
- If $r \leq -1$, $\lim_{n \rightarrow \infty} r^n$ does not exist, because r^n oscillates with an increasing magnitude when $r < -1$, and oscillates between -1 and 1 when $r = -1$.

THEOREMS OF COMPUTING LIMITS - EXAMPLE

$$\text{E.g.: } \lim_{n \rightarrow \infty} \sqrt{\frac{2n+1}{n+3}} = \sqrt{\lim_{n \rightarrow \infty} \frac{2n+1}{n+3}} = \sqrt{\lim_{n \rightarrow \infty} \frac{2 + \frac{1}{n}}{1 + \frac{3}{n}}} = \sqrt{\frac{2+0}{1+0}} = \sqrt{2}.$$

$$\text{E.g.: } \lim_{n \rightarrow \infty} \cos\left(\frac{\pi}{n}\right) = \cos\left(\lim_{n \rightarrow \infty} \frac{\pi}{n}\right) = \cos 0 = 1.$$

E.g.: $\lim_{n \rightarrow \infty} (-3)^n$ does not exist. The sequence $((-3)^n)_{n=1}^{\infty}$ diverges.

$$\text{E.g.: } \lim_{n \rightarrow \infty} \frac{2^{2n}}{5^n} = \lim_{n \rightarrow \infty} \frac{4^n}{5^n} = \lim_{n \rightarrow \infty} \left(\frac{4}{5}\right)^n = 0.$$

$$\text{E.g.: } \lim_{n \rightarrow \infty} (-0.12)^n = 0.$$

$$\text{E.g.: } \lim_{n \rightarrow \infty} (1.0001)^n = \infty.$$

Can you identify the theorems used in each of the examples above?

ACTIVITY 3

(a) Find $\lim_{n \rightarrow \infty} \cos \left(\frac{5n^3 - 2n}{n^4 + 3n^2 - 10} \right)$.

(b) Find $\lim_{n \rightarrow \infty} e^{\left(\frac{\pi^n}{4^n}\right)}$.

(c) In the previous class, we derive the n th partial sum (i.e. the sum of the first n terms) of a geometric sequence to be:

$$S_n := a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a - ar^n}{1 - r} \quad \text{for } r \neq 1,$$

where $a \neq 0$ is the first term and r is the common ratio. What is the infinite sum of the geometric sequence? I.e. what is

$$S_\infty = a + ar + ar^2 + \cdots := \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a - ar^n}{1 - r}?$$

Hint: split into different cases according to $|r| < 1$, $r \leq -1$ and $r > 1$.

ACTIVITY 3: SOLUTION

$$\begin{aligned}
 \text{(a)} \quad \lim_{n \rightarrow \infty} \cos \left(\frac{5n^3 - 2n}{n^4 + 3n^2 - 10} \right) &= \cos \left(\lim_{n \rightarrow \infty} \frac{5n^3 - 2n}{n^4 + 3n^2 - 10} \right) \\
 &= \cos \left(\lim_{n \rightarrow \infty} \frac{\frac{5}{n} - \frac{2}{n^3}}{1 + \frac{3}{n^2} - \frac{10}{n^4}} \right) = \cos \left(\frac{0 - 0}{1 + 0 - 0} \right) = \cos 0 = 1
 \end{aligned}$$

$$\text{(b)} \quad \lim_{n \rightarrow \infty} e^{\left(\frac{\pi}{4}\right)^n} = e^{\lim_{n \rightarrow \infty} \left(\frac{\pi}{4}\right)^n} = e^0 = 1, \text{ since } \left|\frac{\pi}{4}\right| < 1.$$

(c)

$$\lim_{n \rightarrow \infty} \frac{a - ar^n}{1 - r} = \frac{a - a \lim_{n \rightarrow \infty} r^n}{1 - r} = \begin{cases} \frac{a}{1-r} & \text{if } |r| < 1, \\ \infty & \text{if } r > 1, a > 0 \\ -\infty & \text{if } r > 1, a < 0 \\ \text{doesn't exist} & \text{if } r \leq -1. \end{cases}$$

INFINITE SERIES

Informally, a series is an infinite sum:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + a_4 + \cdots$$

However, we can't literally keep adding non-stop, as a non-terminating process doesn't give a definite value. Part (c) of the previous activity shows us how to define series rigorously.

DEFINITION:

A **series** is symbolically the sum of all the terms of a sequence $(a_i)_{i=1}^{\infty}$, i.e.

$$\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + \dots$$

Associated with each series is a sequence of **partial sums** $(S_n)_{n=1}^{\infty}$, where S_n is the **n th partial sum** of the sequence $(a_i)_{i=1}^{\infty}$.

If $\lim_{n \rightarrow \infty} S_n = S \in \mathbb{R}$, then the series is said to **converge** to S , and

S is called the **sum of the series**. We denote this by $\sum_{i=1}^{\infty} a_i = S$. If

the sequence $(S_n)_{n=1}^{\infty}$ diverges, then the series is said to **diverge**.

A divergent series has no sum.

INFINITE SERIES - EXAMPLE

E.g.:

$$\begin{aligned}
 & \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots \\
 &= \sum_{n=1}^{\infty} \frac{1}{2^n} = \lim_{n \rightarrow \infty} S_n \\
 &= \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} = \lim_{n \rightarrow \infty} \left(1 - \left(\frac{1}{2}\right)^n\right) = 1.
 \end{aligned}$$

Here's an intuitive way to see

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1.$$



INFINITE SERIES - EXAMPLE

E.g.: For the series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$, we derived the partial sum in the previous class

$$S_n = 1 - \frac{1}{n+1}.$$

Thus,

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = 1.$$

E.g.: For the series $\sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right)$, we derived the partial sum in the previous class

$$S_n = -\ln(n+1).$$

Thus,

$$\sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right) = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} (-\ln(n+1)) = -\infty,$$

so the series diverges.

INFINITE SERIES

Fun fact: The famous indian mathematician **Srinivasa Ramanujan** discovered that

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{99^2} \sum_{k=0}^{\infty} \frac{(4k)!(1103 + 26390k)}{(k!)^4 396^{4k}},$$

which can be used to compute π efficiently.

Series are also very useful in expressing analytic functions as polynomial of 'infinite degree', called Taylor series. This allows us to numerically compute values of analytic functions with just the four fundamental operations $+$, $-$, $\times$, $\div$. That's how your calculator computes values of $\sqrt[n]{x}$, e^x , $\ln x$, $\sin x$, $\cos x$, $\tan x$, $\arcsin x$, $\arccos x$, $\arctan x$, etc.

DOMINANCE THEOREM

THEOREM

We have the following dominance of growth rates:

$$\text{logarithm} < \text{polynomial} < \text{exponential} < \text{factorial}.$$

More precisely,

- $\lim_{n \rightarrow \infty} \frac{\ln n}{P(n)} = 0$, for any non-constant polynomial function P .
- $\lim_{n \rightarrow \infty} \frac{P(n)}{a^n} = 0$, for any polynomial function P and real number $a > 1$.
- $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$, for any real number a .

This is the reason why $\lim_{n \rightarrow \infty} \frac{2^n}{n^{20}} = \frac{1}{\lim_{n \rightarrow \infty} \frac{n^{20}}{2^n}} = \frac{1}{0^+} = \infty$ in activity

1.

DOMINANCE THEOREM - EXAMPLE

E.g.: $\lim_{n \rightarrow \infty} \frac{n \ln n}{n^2} = \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$. Intuitively, this means $n \ln n$ is arbitrarily small compared to n^2 when n is very large.

$$\text{E.g.: } \lim_{n \rightarrow \infty} \frac{n!}{n^{3000}} = \frac{1}{\lim_{n \rightarrow \infty} \frac{n^{3000}}{n!}} = \frac{1}{\lim_{n \rightarrow \infty} \left(\frac{n^{3000}}{2^n} \frac{2^n}{n!} \right)} = \frac{1}{0^+ \times 0^+} = \infty.$$

Intuitively, this means n^{3000} is arbitrarily small compared to $n!$ when n is very large.

In programming, the **computational complexity** (running time) of an algorithm depends on the size of the input and limits are used to compare different complexities. For instance, to **sort** n numbers (sort them in increasing or decreasing order), algorithm A has complexity $n \ln n$ and algorithm B has complexity n^2 , then algorithm A is preferred as its complexity (running time) is arbitrarily small compared to algorithm B when n is large.

OUTLINE

① Limits of Sequences

② Limits of Functions at Infinity

③ Summary

LIMITS OF FUNCTIONS AT INFINITY

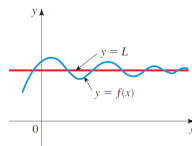
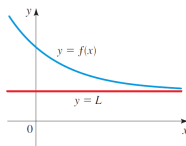
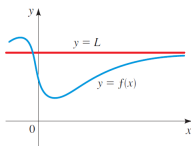
Just like sequences, for functions that are defined over some interval (a, ∞) , we can define in a similar fashion the limit of functions when x gets arbitrarily large.

DEFINITION:

Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to $L \in \mathbb{R}$ by taking x sufficiently large. In this case, we say that $f(x)$ **converges to $L \in \mathbb{R}$ when x tends/approaches to ∞** .



LIMITS OF FUNCTIONS AT NEGATIVE INFINITY

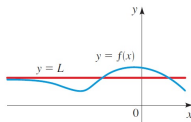
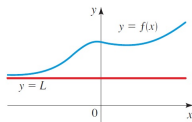
Similarly, for functions that are defined over some interval $(-\infty, a)$, we can define the limit of functions when x gets arbitrarily negatively large.

DEFINITION:

Let f be a function defined on some interval $(-\infty, a)$. Then

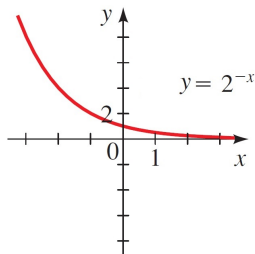
$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to $L \in \mathbb{R}$ by taking x sufficiently negatively large. In this case, we say that $f(x)$ **converges to** $L \in \mathbb{R}$ **when x tends/approaches to** $-\infty$.

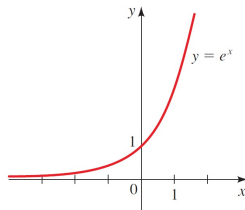


LIMITS OF FUNCTIONS AT ∞ OR $-\infty$ - EXAMPLE

E.g.: $\lim_{x \rightarrow \infty} 2^{-x} = 0.$

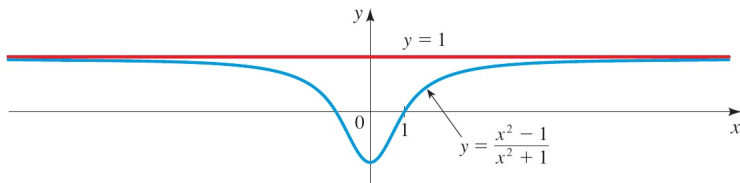


E.g.: $\lim_{x \rightarrow -\infty} e^x = 0.$

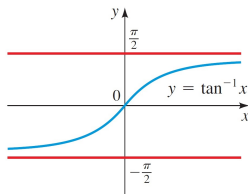


LIMITS OF FUNCTIONS AT ∞ OR $-\infty$ - EXAMPLE

E.g.: $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} = 1$ and $\lim_{x \rightarrow -\infty} \frac{x^2 - 1}{x^2 + 1} = 1$.

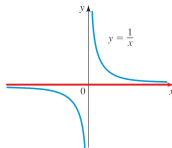


E.g.: $\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$ and $\lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}$.



HORIZONTAL ASYMPTOTES

E.g.: $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ and $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$.



As you can see from the previous examples, graphically, the limits of a function at ∞ or $-\infty$ correspond to the horizontal asymptotes.

DEFINITION:

The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

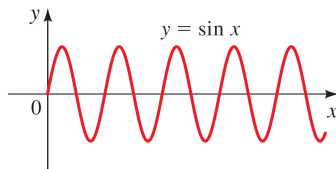
$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L.$$

DIVERGENCE

DEFINITION:

If $f(x)$ does not converge to **any** $L \in \mathbb{R}$ when x approaches ∞ (resp. $-\infty$), we say that $\lim_{x \rightarrow \infty} f(x)$ (resp. $\lim_{x \rightarrow -\infty} f(x)$) **doesn't exist** or $f(x)$ **diverges** when x approaches ∞ (resp. $-\infty$).

E.g.: Both $\lim_{x \rightarrow \infty} \sin x$ and $\lim_{x \rightarrow -\infty} \sin x$ do not exist.



RULES OF COMPUTING LIMITS

THEOREM

Suppose $\lim_{x \rightarrow \infty} f(x) = L_1$ and $\lim_{x \rightarrow \infty} g(x) = L_2$ (L_1, L_2 can be $\pm\infty$) and let $c \in \mathbb{R}$ be a constant. Then

- $\lim_{x \rightarrow \infty} c = c,$
- $\lim_{x \rightarrow \infty} cf(x) = c \left(\lim_{x \rightarrow \infty} f(x) \right) = cL_1,$
- $\lim_{x \rightarrow \infty} (f(x) \pm g(x)) = \lim_{x \rightarrow \infty} f(x) \pm \lim_{x \rightarrow \infty} g(x) = L_1 \pm L_2,$
- $\lim_{x \rightarrow \infty} (f(x) \cdot g(x)) = \lim_{x \rightarrow \infty} f(x) \times \lim_{x \rightarrow \infty} g(x) = L_1 \cdot L_2,$
- $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow \infty} f(x)}{\lim_{x \rightarrow \infty} g(x)} = \frac{L_1}{L_2},$

with the same conventions and indeterminate forms as limits of sequences.

END BEHAVIOR AND HORIZONTAL ASYMPTOTES OF POLYNOMIAL AND RATIONAL FUNCTIONS

The rules stated in the previous slide are also applicable when $x \rightarrow -\infty$. You might notice that the rules are identical to the ones for sequences.

These rules of limits can be used to explain all the end behavior and horizontal asymptotes of polynomial and rational functions that we have gone through in Week 2.

E.g.: If $P(x) = -2x^4 + 5x^3 + 4x - 7$, then

$$\begin{aligned}\lim_{x \rightarrow \infty} P(x) &= \lim_{x \rightarrow \infty} (-2x^4 + 5x^3 + 4x - 7) \\ &= \lim_{x \rightarrow \infty} x^4 \left(-2 + \frac{5}{x} + \frac{4}{x^3} - \frac{7}{x^4} \right) \\ &= \lim_{x \rightarrow \infty} x^4 \cdot \lim_{x \rightarrow \infty} \left(-2 + \frac{5}{x} + \frac{4}{x^3} - \frac{7}{x^4} \right) \\ &= \infty \times (-2 + 0 + 0 - 0) = -\infty.\end{aligned}$$

END BEHAVIOR AND HORIZONTAL ASYMPTOTES OF POLYNOMIAL AND RATIONAL FUNCTIONS - EXAMPLE

E.g.: If $P(x) = -2x^4 + 5x^3 + 4x - 7$, then

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} P(x) &= \lim_{x \rightarrow -\infty} (-2x^4 + 5x^3 + 4x - 7) \\
 &= \lim_{x \rightarrow -\infty} x^4 \left(-2 + \frac{5}{x} + \frac{4}{x^3} - \frac{7}{x^4} \right) \\
 &= \lim_{x \rightarrow -\infty} x^4 \cdot \lim_{x \rightarrow -\infty} \left(-2 + \frac{5}{x} + \frac{4}{x^3} - \frac{7}{x^4} \right) \\
 &= (-\infty) \times (-\infty) \times (-\infty) \times (-\infty) \times (-2 + 0 + 0 - 0) \\
 &= \infty \times (-2) = -\infty.
 \end{aligned}$$

END BEHAVIOR AND HORIZONTAL ASYMPTOTES OF POLYNOMIAL AND RATIONAL FUNCTIONS - EXAMPLE

E.g.: If $R(x) = \frac{8x^2 + 1}{4x^2 + 2x - 6}$, then

$$\begin{aligned}\lim_{x \rightarrow \infty} R(x) &= \lim_{x \rightarrow \infty} \frac{8x^2 + 1}{4x^2 + 2x - 6} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 \left(8 + \frac{1}{x^2}\right)}{x^2 \left(4 + \frac{2}{x} - \frac{6}{x^2}\right)} \\ &= \frac{\lim_{x \rightarrow \infty} \left(8 + \frac{1}{x^2}\right)}{\lim_{x \rightarrow \infty} \left(4 + \frac{2}{x} - \frac{6}{x^2}\right)} \\ &= \frac{8 + 0}{4 + 0 - 0} = 2.\end{aligned}$$

Similar working gives $\lim_{x \rightarrow -\infty} R(x) = 2$.

ACTIVITY 4

Find the limits of the following functions when x approaches ∞ or $-\infty$, if they exist.

(a) $P(x) = x^4 + x^3 - 16x^2 - 4x + 48$

(b) $Q(x) = -x(x - 3)(x + 2)$

(c) $R(x) = -(x + 1)^2(x - 1)^3(x - 2)$

(d) $f(x) = \frac{2x - 3}{x^2 - 1}$

(e) $g(x) = \frac{x^3 + 3x^2}{x^2 - 4}$

ACTIVITY 4: SOLUTION

$$\begin{aligned}
 \text{(a)} \quad \lim_{x \rightarrow \infty} P(x) &= \lim_{x \rightarrow \infty} (x^4 + x^3 - 16x^2 - 4x + 48) \\
 &= \lim_{x \rightarrow \infty} x^4 \left(1 + \frac{1}{x} - \frac{16}{x^2} - \frac{4}{x^3} + \frac{48}{x^4} \right) \\
 &= \lim_{x \rightarrow \infty} x^4 \cdot \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} - \frac{16}{x^2} - \frac{4}{x^3} + \frac{48}{x^4} \right) \\
 &= \infty \times (1 + 0 - 0 - 0 + 0) = \infty.
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} P(x) &= \lim_{x \rightarrow -\infty} (x^4 + x^3 - 16x^2 - 4x + 48) \\
 &= \lim_{x \rightarrow -\infty} x^4 \left(1 + \frac{1}{x} - \frac{16}{x^2} - \frac{4}{x^3} + \frac{48}{x^4} \right) \\
 &= \lim_{x \rightarrow -\infty} x^4 \cdot \lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x} - \frac{16}{x^2} - \frac{4}{x^3} + \frac{48}{x^4} \right) \\
 &= (-\infty)^4 \times (1 + 0 - 0 - 0 + 0) = \infty.
 \end{aligned}$$

Both $\lim_{x \rightarrow \infty} P(x)$ and $\lim_{x \rightarrow -\infty} P(x)$ diverge to ∞ .

ACTIVITY 4: SOLUTION

(b)

$$\begin{aligned}
 \lim_{x \rightarrow \infty} Q(x) &= \lim_{x \rightarrow \infty} (-x(x-3)(x+2)) \\
 &= \lim_{x \rightarrow \infty} (-x) \cdot \lim_{x \rightarrow \infty} (x-3) \cdot \lim_{x \rightarrow \infty} (x+2) \\
 &= (-\infty) \cdot \infty \cdot \infty \\
 &= -\infty.
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} Q(x) &= \lim_{x \rightarrow -\infty} (-x(x-3)(x+2)) \\
 &= \lim_{x \rightarrow -\infty} (-x) \cdot \lim_{x \rightarrow -\infty} (x-3) \cdot \lim_{x \rightarrow -\infty} (x+2) \\
 &= \infty \cdot (-\infty) \cdot (-\infty) \\
 &= \infty.
 \end{aligned}$$

$\lim_{x \rightarrow \infty} Q(x)$ diverges to $-\infty$ and $\lim_{x \rightarrow -\infty} Q(x)$ diverges to ∞ .

ACTIVITY 4: SOLUTION

(c)

$$\begin{aligned}
 \lim_{x \rightarrow \infty} R(x) &= \lim_{x \rightarrow \infty} \left(-(x+1)^2(x-1)^3(x-2) \right) \\
 &= \lim_{x \rightarrow \infty} \left(-(x+1)^2 \right) \cdot \lim_{x \rightarrow \infty} (x-1)^3 \cdot \lim_{x \rightarrow \infty} (x-2) \\
 &= -(\infty)^2 \cdot (\infty)^3 \cdot \infty \\
 &= -\infty.
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} R(x) &= \lim_{x \rightarrow -\infty} \left(-(x+1)^2(x-1)^3(x-2) \right) \\
 &= \lim_{x \rightarrow -\infty} \left(-(x+1)^2 \right) \cdot \lim_{x \rightarrow -\infty} (x-1)^3 \cdot \lim_{x \rightarrow -\infty} (x-2) \\
 &= -(-\infty)^2 \cdot (-\infty)^3 \cdot (-\infty) \\
 &= (-\infty) \cdot (-\infty) \cdot (-\infty) = -\infty.
 \end{aligned}$$

Both $\lim_{x \rightarrow \infty} R(x)$ and $\lim_{x \rightarrow -\infty} R(x)$ diverges to $-\infty$.

ACTIVITY 4: SOLUTION

$$\begin{aligned}
 \text{(d)} \quad \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{2x - 3}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{x^2 \left(\frac{2}{x} - \frac{3}{x^2} \right)}{x^2 \left(1 - \frac{1}{x^2} \right)} \\
 &= \lim_{x \rightarrow \infty} \frac{\left(\frac{2}{x} - \frac{3}{x^2} \right)}{\left(1 - \frac{1}{x^2} \right)} = \frac{\lim_{x \rightarrow \infty} \left(\frac{2}{x} - \frac{3}{x^2} \right)}{\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x^2} \right)} \\
 &= \frac{0 - 0}{1 - 0} = \frac{0}{1} = 0.
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{2x - 3}{x^2 - 1} = \lim_{x \rightarrow -\infty} \frac{x^2 \left(\frac{2}{x} - \frac{3}{x^2} \right)}{x^2 \left(1 - \frac{1}{x^2} \right)} \\
 &= \lim_{x \rightarrow -\infty} \frac{\left(\frac{2}{x} - \frac{3}{x^2} \right)}{\left(1 - \frac{1}{x^2} \right)} = \frac{\lim_{x \rightarrow -\infty} \left(\frac{2}{x} - \frac{3}{x^2} \right)}{\lim_{x \rightarrow -\infty} \left(1 - \frac{1}{x^2} \right)} \\
 &= \frac{0 - 0}{1 - 0} = \frac{0}{1} = 0.
 \end{aligned}$$

ACTIVITY 4: SOLUTION

$$\begin{aligned}
 \text{(e)} \quad \lim_{x \rightarrow \infty} g(x) &= \lim_{x \rightarrow \infty} \frac{x^3 + 3x^2}{x^2 - 4} = \lim_{x \rightarrow \infty} \frac{x^2(x + 3)}{x^2 \left(1 - \frac{4}{x^2}\right)} \\
 &= \lim_{x \rightarrow \infty} \frac{(x + 3)}{\left(1 - \frac{4}{x^2}\right)} = \frac{\lim_{x \rightarrow \infty} (x + 3)}{\lim_{x \rightarrow \infty} \left(1 - \frac{4}{x^2}\right)} \\
 &= \frac{\infty}{1 - 0} = \infty.
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} g(x) &= \lim_{x \rightarrow -\infty} \frac{x^3 + 3x^2}{x^2 - 4} = \lim_{x \rightarrow -\infty} \frac{x^2(x + 3)}{x^2 \left(1 - \frac{4}{x^2}\right)} \\
 &= \lim_{x \rightarrow -\infty} \frac{(x + 3)}{\left(1 - \frac{4}{x^2}\right)} = \frac{\lim_{x \rightarrow -\infty} (x + 3)}{\lim_{x \rightarrow -\infty} \left(1 - \frac{4}{x^2}\right)} \\
 &= \frac{-\infty}{1 - 0} = -\infty.
 \end{aligned}$$

$\lim_{x \rightarrow \infty} g(x)$ diverges to ∞ and $\lim_{x \rightarrow -\infty} g(x)$ diverges to $-\infty$.

OUTLINE

① Limits of Sequences

② Limits of Functions at Infinity

③ Summary

SUMMARY

- Limits of Sequences
 - definition of the limit of a sequence.
 - convergent and divergent sequences.
 - rules of computing limits.
 - theorems of computing limits.
 - infinite series.
 - Dominance Theorem.
- Limits of Functions at Infinity
 - definition of limits of functions at ∞ and $-\infty$.
 - horizontal asymptotes.
 - divergence and infinity divergence.
 - rules of computing limits.
 - end behavior and horizontal asymptotes of polynomial and rational functions

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) section 13.4.